Inference & Bayesian networks

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11.1 Inference and Uncertainty

"As far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality."

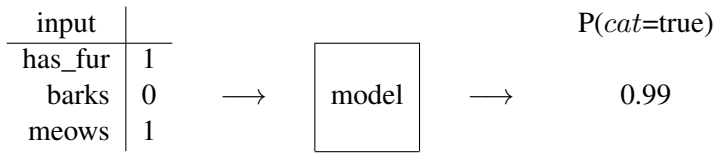
Albert Einstein, 1921

Because we need to

quantify our degree of belief/

quantify our uncertainty

11.1.1 The probabilistic model



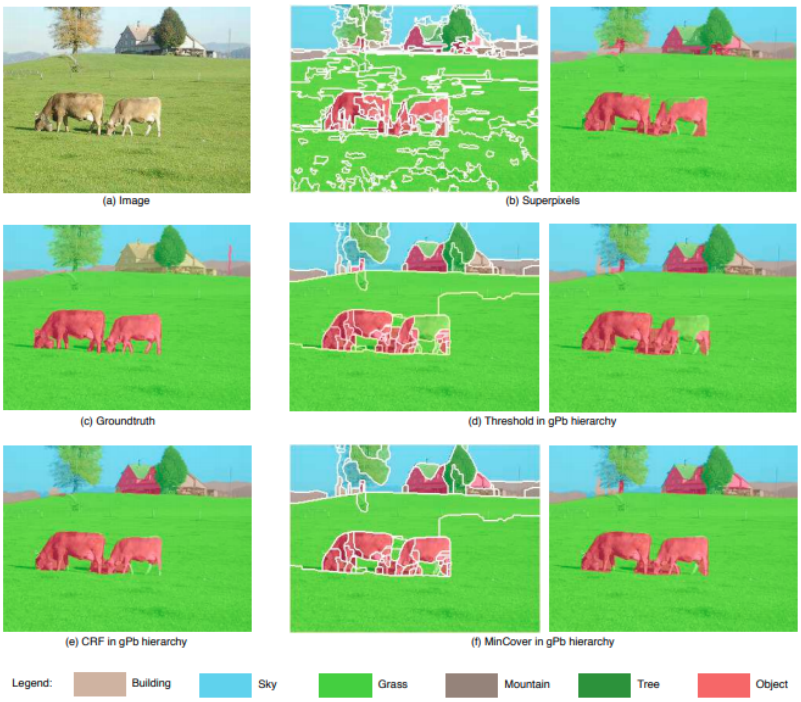


Figure 11.1.1: Semantic segmentation (pixel-level classification) (Farabet et al., 2012)

- <u>the model:</u>

$$P(H) : H \to [0,1] \tag{11.1.1}$$
$$P(H) = 0 \quad \iff \quad H \text{ is false (i.e. impossible)} \tag{11.1.2}$$
$$P(H) = 1 \quad \iff \quad H \text{ is true (i.e. certain)} \tag{11.1.3}$$
$$0 < P(H) < 1 \quad \text{quantifies our degree of belief} \tag{11.1.4}$$

- <u>events</u>: assignment to a set of cases (e.g. two dice add up to 11)
- <u>proposition</u>: disjunction of **events** (i.e. isolate the sets of **events** in which the **proposition** holds)
  - e.g. $cavity = \text{true}$ is equivalent to: $(cavity = \text{true} \wedge toothache = \text{false} \wedge catch = \text{false}) \vee (cavity = \text{true} \wedge toothache = \text{true} \wedge catch = \text{true}) \vee \ldots$

$\underbrace{\textit{Summing over}}_{\text{marginalization}}$ the probabilities of the **events** $\rightsquigarrow$ probability of **proposition**.

Example with dice:

$$\overbrace{P(total = 11)}^{\text{A}} = \overbrace{P(die_1 = 5, die_2 = 6)}^{\text{B}} + \overbrace{P(die_1 = 6, die_2 = 5)}^{\text{C}} \tag{11.1.5}$$
$$= 1/36 + 1/36 \tag{11.1.6}$$
$$= 1/18 \tag{11.1.7}$$

The terms A,B and C are *unconditional* or *prior* probabilities. They represent the degree of belief in the absence of any other information. Priors are also referred to as *domain knowledge*.

11.1.2 Evidence

What if we already *know* (i.e. observed) something?

e.g. We've already rolled the first die and got 5. Now we're waiting to roll the second die.

We are now interested in the *conditional* (or *posterior*) probability of the second die leading to a total of 11:

$$P(\ total = 11 \underbrace{\mid}_{\text{"given"}} \overbrace{die_1}^{\{1,2,3,4,5,6\}} = 5\ ) \tag{11.1.8}$$

The prior is valid even after the outcome.

e.g.

$$P(\overbrace{cavity}^{\substack{\text{shorthand for}\\ cavity=\text{true}}}) = 0.2$$

$$P(cavity|\ \overbrace{toothache}^{toothache=\text{true}}\ ) = 0.6$$

If we observe that a $cavity$ has actually occurred, we don't have to change our prior on cavities. However, the prior becomes less useful if we proceed to infer other things.

Caution in interpreting conditional probabilities:

- × Whenever $toothache$ is true, then $cavity$ is true with probability of 0.6
- ✓ Whenever $toothache$ is true, <u>and no other information</u> is available, then $cavity$ is true with probability of 0.6.

other information: e.g. diagnosis was <u>no</u> $cavity$. In this case, the probability is no longer 0.6, but:

$$P(cavity \mid toothache \wedge diagnosis_{cavity} = \text{false}) \overset{\text{e.g.}}{=} 0.02 \tag{11.1.9}$$

Observing event e rules out events where e is false. This leaves a set where $P(e) > 0$. Therefore, when we query for some variable x, we look within that set for the fraction that satisfies x and e.

This fraction can be calculated using the following:

$$P(x|e) = \frac{P(x,e)}{P(e)} \tag{11.1.10}$$

which brings us to the ***product rule***:

$$P(x,e) = P(x|e)P(e) \tag{11.1.11}$$

Q 11.1.1: *What information do I need to calculate the probability of any proposition?*

- The full joint probability distribution provides all the information to calculate the probability of any proposition.

Example of a full joint distribution between $Cavity, Toothache$ and $Catch$:

	$toothache =$ true		$toothache =$ false	
	$catch =$ true	$catch =$ false	$catch =$ true	$catch =$ false
$cavity =$ true	0.108	0.012	0.072	0.008
$cavity =$ false	0.016	0.064	0.144	0.576

11.1.2.1 Marginalization ("summing out")

$$P(x|\underline{\mathbf{e}}) = \frac{P(x, \underline{\mathbf{e}})}{P(\underline{\mathbf{e}})} = \alpha P(x, \underline{\mathbf{e}}) = \alpha \sum_{\underline{\mathbf{y}} \in \underline{\mathbf{Y}}} P(\underline{\mathbf{x}}, \underline{\mathbf{e}}, \underline{\mathbf{y}}) \tag{11.1.12}$$

where

x: query variable

$\underline{\mathbf{e}}$: evidence variable(s)

$\underline{\mathbf{y}}$: **un**observed variables in the table

α: normalization factor:

Q 11.1.2: *How do we calculate the normalization factor α?*

- Via:

1. evidence prior:

$$\frac{1}{\alpha} := P(\underline{\mathbf{e}}) = \sum_{x, \underline{\mathbf{y}}} P(x, \underline{\mathbf{e}}, \underline{\mathbf{y}}) \tag{11.1.13}$$

2. or simply whatever ensures $\sum_x P(x, \underline{\mathbf{e}}) \stackrel{!}{=} 1$

11.1.2.2 Applying the product rule

$$P(cavity|toothache = \text{true}) = \frac{P(cavity, toothache)}{P(toothache)} \tag{11.1.14}$$

$$= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064} \tag{11.1.15}$$

$$= 0.6 \tag{11.1.16}$$

$$P(cavity = \text{false}|toothache = \text{true}) = ?$$

Q 11.1.3: *What should we expect?*

$$P(cavity = \text{false}|toothache = \text{true}) = \frac{P(cavity = \text{false}, toothache)}{P(toothache)} \tag{11.1.17}$$

$$= \frac{0.016 + 0.0064}{0.108 + 0.012 + 0.016 + 0.064} \tag{11.1.18}$$

$$= 0.4 = 1 - 0.6 \tag{11.1.19}$$

$$\begin{aligned}
P(\overbrace{Cavity}^{\begin{pmatrix} cavity = \text{true} \\ cavity = \text{false} \end{pmatrix}} | toothache) &= \alpha P(Cavity, toothache) && (11.1.20)\\
&= \alpha\big[P(Cavity, toothache, catch) && (11.1.21)\\
&\quad + P(Cavity, toothache, catch = \text{false})\big] && (11.1.22)\\
&= \alpha\left[\begin{pmatrix} 0.108 \\ 0.016 \end{pmatrix} + \begin{pmatrix} 0.012 \\ 0.064 \end{pmatrix}\right] && (11.1.23)\\
&= \alpha \begin{pmatrix} 0.12 \\ 0.08 \end{pmatrix} \overset{\alpha = 1/P(toothache)}{=} \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} && (11.1.24)
\end{aligned}$$

OR normalize s.t. $\sum_{\underline{\mathbf{x}}} P(x, \underline{\mathbf{e}}) \overset{!}{=} 1$

$$= \alpha \begin{pmatrix} 0.12 \\ 0.08 \end{pmatrix} \overset{\alpha = \frac{1}{0.12+0.08}}{=} \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} \qquad (11.1.25)$$

11.2 Independence

11.2.1 Motivation

The full joint distribution has everything we need to perform inference, but it scales badly with more and more variables (e.g. adding $Weather$ (e.g. # of weather conditions = 4) leads to a table with $2 \times 2 \times 2 \times 4 = 32$ entries)

Q 11.2.1: *Should the weather be influenced by a $cavity$?*

$$P(cloudy | toothache, catch, cavity) \overset{!}{=} P(cloudy) \qquad (11.2.1)$$

Q 11.2.2: *Should the weather have any influence on cavities, $toothache$ or the dentist?*

-No, this independence can be formulated by:

$$\begin{aligned}
&P(toothache, catch, cavity, Weather) && (11.2.2)\\
&\overset{!}{=} P(toothache, catch, cavity) P(Weather) && (11.2.3)
\end{aligned}$$

The 32 elements in the table can be split into the original 8 + 4 (new table with 4 entries for the weather).

"Extreme" case: n independent coin flips: 2^n combinations. Independence allows us to reduce this to $n\times$ single-variable distributions.

11.2.2 Bayes' theorem

Start with the product rule:

$$P(X|Y)P(Y) = P(Y,X) = P(X,Y) = P(Y|X)P(X) \tag{11.2.4}$$

$$P(X|Y)P(Y) = P(Y|X)P(X) \tag{11.2.5}$$

solve for $P(Y|X)$ to get Bayes' rule

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)} \tag{11.2.6}$$

How to read Bayes' theorem:

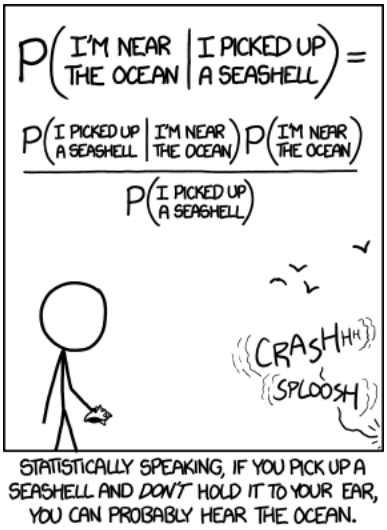


Figure 11.2.1: Bayes' Theorem example from xkcd.com

As a *cause* and *effect* relationship:

$$\underbrace{P(cause|effect)}_{\text{diagnositc}} = \frac{\overbrace{P(effect|cause)}^{\text{causal}}\,P(cause)}{P(effect)} \tag{11.2.7}$$

If $P(X)$ is missing, we can still normalize using:

$$P(Y|X) = \alpha P(X|Y)P(Y) \tag{11.2.8}$$

α can be chosen to whatever will make the entries in $P(Y|X)$ sum up to 1.

11.2.3 Using Bayes' theorem

$$P(Y|X) = \alpha P(X|Y)P(Y) \tag{11.2.9}$$

$$P(Cavity|toothache = \text{true}, catch = \text{true}) \tag{11.2.10}$$
$$= \alpha \underbrace{P(toothache = \text{true}, catch = \text{true}|Cavity)}_{\circledast} P(Cavity) \tag{11.2.11}$$

$\circledast$ similar scaling problem, not much better than the case of using the full joint distribution.

Q 11.2.3: *Can we use independence to further mitigate the scaling problem here?*

$Toothache$ and $Catch$ are not independent but $Toothache$ given $cavity$ is independent of $Catch$ given $cavity$.

- $toothache$ depends on $cavity$: nerves & tolerance for pain.
- $catch$ depends on $cavity$: dependent on how skillful the dentist is.
- **But**... "nerves & tolerance for pain" is <u>independent</u> of "how skillful the dentist is".

Therefore:

$$P(toothache, catch|Cavity) \tag{11.2.12}$$
$$= P(toothache|Cavity)P(catch|Cavity) \tag{11.2.13}$$

i.e. the *conditional* independence of $toothache$ and $catch$ <u>given</u> $Cavity$

Plugging Eq.(11.2.13) into Eq.(11.2.11) yields:

$$P(Cavity|toothache, catch) \tag{11.2.14}$$
$$= \alpha P(toothache|Cavity)P(catch|Cavity)P(Cavity) \tag{11.2.15}$$

11.2.4 The Naive Bayes model

An alternate expression for the full joint distribution by exploiting conditional independence.

We start with the product rule:

$$P(X, Y) = P(X|Y)P(Y) \tag{11.2.16}$$

$$P(Toothache, Catch, Cavity) = P(Toothache, Catch|Cavity)\underbrace{P(Cavity)}_{\text{prior}} \tag{11.2.17}$$

Exploiting conditional independence yields:

$$P(Toothache, Catch, Cavity) \tag{11.2.18}$$
$$= P(Toothache|Cavity)P(Catch|Cavity)\underbrace{P(Cavity)}_{\text{prior}} \tag{11.2.19}$$

In terms of cause and effect:

$$P(Cause, Effect_1, Effect_2, \ldots, Effect_n) \tag{11.2.20}$$
$$= \underbrace{P(Cause)}_{\text{prior}} \prod_{i=1}^{n} P(Effect_i|Cause) \tag{11.2.21}$$

Complexity reduced!

Eq.(11.2.21) is the formulation of the *Naive Bayes model*.

11.3 Bayesian Networks (part 1/2)

11.3.1 Directed Acyclic Graphs (DAG)

A graphical representation of conditional independence:

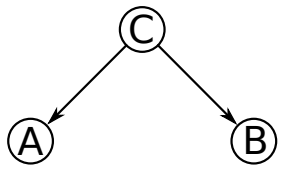


Figure 11.3.1: The conditional independence between A and B given C.

This leads to the factorization of the full joint distribution:

$$P(A, B, C) = P(A|C)P(B|C)P(C) \tag{11.3.1}$$

Q 11.3.1: *What does the DAG representation of the full joint distribution look like for toothache example?*

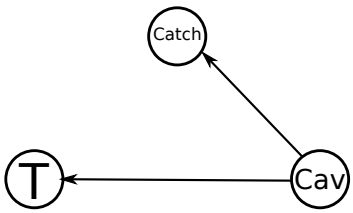
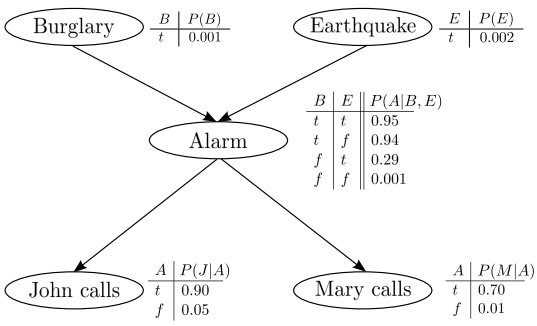


Figure 11.3.2: DAG representation for the toothache example.

11.3.1.1 A "Californian" example



Considering statistical dependencies, only 10 instead of 31 entries have to be stored.

$$P(J, M, A, B, E) \overset{\text{product rule}}{=} P(J|M, A, B, E)\, P(M|A, B, E)\, P(A|B, E)\, P(B|E)\, P(E) \tag{11.3.2}$$

$$\overset{\text{cond. indep.}}{=} P(J|A)\, P(M|A)\, P(A|B, E)\, P(B)\, P(E) \tag{11.3.3}$$

A factorization of the full joint distribution[1]:

[1] See Ch. 14 from (Russell and Norvig, 2016) for more details.

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i | parents(X_i)) \tag{11.3.4}$$

Remember: independence is important. Independence can be identified via the Markov blanket

The Markov blanket of a node includes:

- its parents
- its children
- other parents of the children

11.3.2 Preparations to perform inference

1. use topological representation to ensure a compact representation. The purpose of topological sorting is to identify cliques in the chordal decomposable graph (next week)
2. turn DAG into a *moral graph* i.e. an undirected graph + new edges s.t. each node of the original DAG is now directly connected to the nodes of its *Markov blanket*
3. Perform inference efficiently (next week) by applying *message passing* on the *Junction tree*

11.3.2.1 Topological sorting

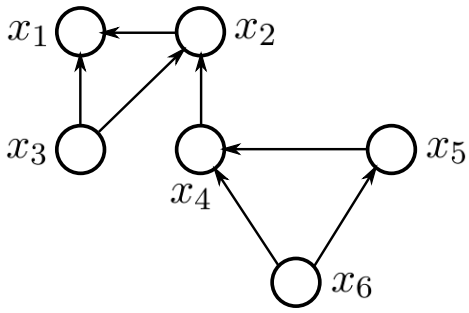
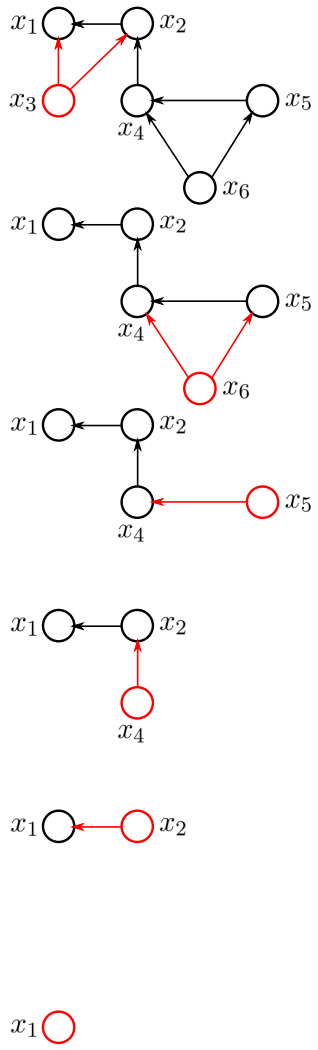


Figure 11.3.3: Example DAG

- Edges should always be directed from nodes with lower to nodes with higher indices.

- Topological sorting: From parents to children:
  1. select and queue a node without parents
  2. delete that node from the DAG
  3. repeat until DAG is empty
  4. renumber nodes



- example:
  - $x_3, x_6, x_5, x_4, x_2, x_1$
- other possible topological orderings:
  - $x_6, x_5, x_4, x_3, x_2, x_1$
  - $x_6, x_5, x_3, x_4, x_2, x_1$
  - $x_6, x_3, x_5, x_4, x_2, x_1$

Exercise: topological sorting

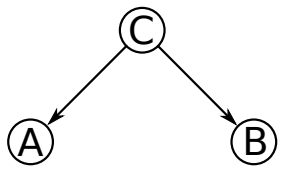


Figure 11.3.4: DAG

- Topological sorting: From parents to children:
  1. select and queue a node without parents
  2. delete that node from the DAG
  3. repeat until DAG is empty
  4. renumber nodes

(see blackboard)

Exercise: topological sorting

(see blackboard)

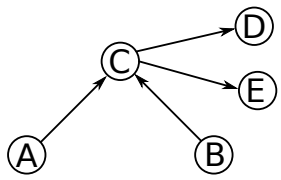


Figure 11.3.5: DAG

Exercise: DAG (same as before) to moral graph

(see blackboard)

11.4 References

References

Clement Farabet, Camille Couprie, Laurent Najman, and Yann LeCun. Learning hierarchical features for scene labeling. *IEEE transactions on pattern analysis and machine intelligence*, 35(8):1915–1929, 2012.

Stuart J Russell and Peter Norvig. *Artificial intelligence: a modern approach*. Malaysia; Pearson Education Limited,, 2016.